

AIRLINE CUSTOMER INTELLIGENCE

Understanding What Drives Passenger Recommendation

Prepared by

Aqeel Shahzad

Dataset

21,788 Airline Reviews

Data & Analytics

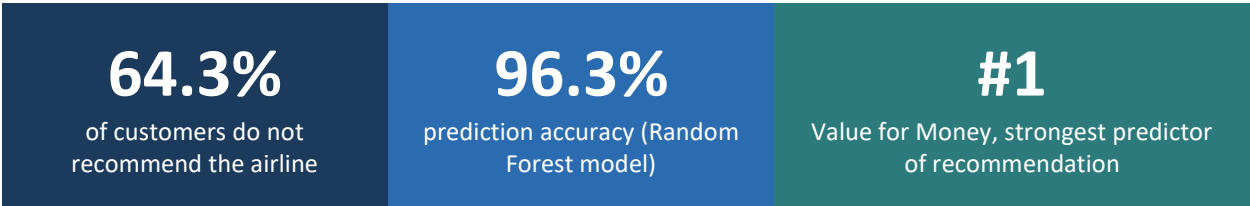
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KEY FINDING

Perceived value and operational reliability are the primary drivers of customer recommendation, not premium services or onboard extras.

EXECUTIVE SUMMARY

This report presents findings from a business analytics case study of 21,788 airline customer reviews, combining structured ratings with passenger-written text to identify the key drivers of recommendation behaviour. The analysis was conducted using machine learning methods including predictive modelling, sentiment analysis, and text classification.



Three strategic priorities emerge from this analysis:

- ▶ Prioritise perceived value, it is the single strongest predictor of whether a customer recommends the airline.
- ▶ Eliminate operational failures, delays, cancellations, refund failures, and poor customer support are strongly linked to non-recommendation, with far greater impact than incremental service improvements.
- ▶ Manage the customer journey holistically, recommendation is outcome-driven; a single major failure outweighs multiple positive service moments.

ABOUT THIS ANALYSIS

This analysis examines what determines whether an airline passenger chooses to recommend that airline. Data from 21,788 verified customer reviews was analysed across two dimensions: structured numerical service ratings (seat comfort, cabin staff, food, value for money, etc.) and free-text written reviews. Both structured and unstructured data were modelled independently and in combination.

Area	Detail
Dataset	21,788 airline customer reviews (post-cleaning)
Review period	2005 – 2023
Outcome variable	Customer recommendation (Yes / No)
Analytical methods	Logistic Regression, Random Forest, Linear SVM, VADER and RoBERTa Sentiment Analysis, TF-IDF Text Modelling
Validation approach	Random split and time-based split testing

CURRENT STATE OF CUSTOMER SATISFACTION

Customer satisfaction across this dataset is low. The average overall rating is 3.14 out of 10, with a median of just 1, meaning the majority of reviews are highly negative. This skew is reflected in recommendation behaviour: 64.3% of passengers do not recommend the airline.

Service Ratings at a Glance

Service Dimension	Mean Score	Out of	Business Implication
Value for Money	2.47	5	Lowest perceived value , biggest recommendation lever
Cabin Staff Service	2.90	5	Above average but still below expectations
Seat Comfort	2.68	5	Moderate , premium passengers particularly affected
Ground Service	2.35	5	Weak , a notable pain point on the ground
Food & Beverages	1.69	5	Below average , highly visible to passengers
Inflight Entertainment	1.08	5	Very low , largely unavailable or not rated
Wifi & Connectivity	0.48	5	Near zero , minimal availability across fleet
Overall Rating	3.14	10	Low , driven by disproportionate negative reviews

A notable finding is that 13.6% of reviews show inconsistency between overall rating and recommendation, passengers who gave a high numerical rating did not recommend the airline, or vice versa. This indicates that recommendation is not simply a reflection of aggregate scores: specific service encounters, particularly operational failures, play a decisive role.

WHAT DRIVES CUSTOMER RECOMMENDATION

Analysis of the relationship between service ratings and recommendation behaviour reveals a clear threshold effect: recommendation does not increase gradually with service quality. Instead, it remains low across most rating levels before rising sharply once passengers perceive sufficient value. This has significant strategic implications.

KEY INSIGHT

Recommendation rates remain near zero for passengers rating Value for Money at 1–3, then rise steeply to over 90% for passengers rating it at 4–5. Getting customers past this threshold is far more impactful than refining already-adequate service.

Correlation with Recommendation

The following factors, ranked by strength of association with recommendation, guide where investment will have the most impact:

#	Factor	Correlation	What This Means
1	Value for Money	Very High	The dominant driver , pricing and expectation alignment are critical
2	Overall Rating	Very High	A proxy for total experience; must be managed holistically
3	Cabin Staff Service	Moderate	Friendly, helpful staff meaningfully improve recommendation rates
4	Seat Comfort	Moderate	Matters most in economy and premium economy travel
5	Food & Beverages	Moderate	Visible quality signal, especially on longer routes
6	Ground Service	Lower	Important but less decisive than inflight experience
7	Wifi & Connectivity	Minimal	Low availability limits its impact across the dataset

THE OUTSIZED IMPACT OF OPERATIONAL FAILURES

Text analysis of passenger reviews reveals that operational failures , not incremental service quality , are the primary cause of non-recommendation. Passengers who write about delays, cancellations, refunds, or customer support are dramatically less likely to recommend the airline, regardless of how they rate other service dimensions.

Recommendation Rate by Complaint Theme

Complaint Theme in Review	Not Rec. (%)	Rec. (%)	Priority Level
Customer Support Issues	41.6%	10.0%	CRITICAL , 31.5pp gap
Cancellations	21.6%	2.5%	CRITICAL , 19.1pp gap
Refund Problems	17.2%	1.5%	CRITICAL , 15.7pp gap
Delays	34.7%	19.5%	HIGH , 15.2pp gap
Baggage Issues	29.7%	23.0%	HIGH , 6.8pp gap
Boarding Issues	36.4%	38.2%	NEUTRAL , marginal difference
Seat Experience	26.3%	43.3%	POSITIVE , more common in recommendations
Staff Experience	59.3%	80.5%	POSITIVE , stronger driver when positive
Food & Beverage	21.3%	45.8%	POSITIVE , 24.5pp more common in recommendations

IMPLICATION	Preventing a customer from experiencing a cancellation or refund failure has a greater positive effect on recommendation than delivering excellent food and beverage service. Operational resilience is not just a cost management issue , it is a direct driver of customer advocacy.
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PREDICTIVE MODEL PERFORMANCE

Machine learning models were trained to predict whether a passenger will recommend the airline. This enables the business to identify at-risk customers, prioritise service recovery interventions, and segment communication strategies.

Model Accuracy Summary

Model	Accuracy	Precision	Recall	ROC-AUC
Numerical Random Forest ★	96.3%	95.3%	94.3%	98.8%
Combined Linear SVM	96.2%	95.4%	93.8%	98.7%
Combined Random Forest	96.1%	95.3%	93.8%	98.8%
Numerical Logistic Regression	94.5%	92.9%	91.6%	97.9%
Text Linear SVM (Text Only)	94.1%	93.2%	90.0%	97.8%

The Random Forest model using structured numerical ratings achieves the strongest predictive performance with 96.3% accuracy and a ROC-AUC of 0.988, indicating very strong separation between customers who recommend and do not recommend. Combining text features with structured ratings does not significantly improve accuracy, but text analysis adds diagnostic value by identifying the specific service failures driving dissatisfaction.

WHAT CUSTOMERS ARE SAYING

Text analysis of review language identifies the words and phrases most strongly associated with recommendation and non-recommendation. These patterns support targeted service improvement and communication strategy.

LANGUAGE IN POSITIVE REVIEWS	LANGUAGE IN NEGATIVE REVIEWS
✓ friendly	✗ flight delayed
✓ excellent service	✗ cancelled
✓ comfortable seat	✗ no refund
✓ great staff	✗ would not recommend
✓ pleasant experience	✗ terrible experience
✓ smooth flight	✗ poor customer service
✓ helpful crew	✗ never again
✓ clean aircraft	✗ disappointed
✓ good food	✗ worst airline
✓ impressed	✗ cannot

STRATEGIC RECOMMENDATIONS

Based on the analytical findings, three strategic priorities are recommended. These are ordered by expected impact on customer recommendation rates.

01

Align Pricing with Customer Value Expectations

Value for Money is the single strongest predictor of customer recommendation. Customers who rate value at 4–5 recommend at rates exceeding 90%; those rating it at 1–3 rarely recommend at all. Pricing strategy, bundle design, and expectation management at point of booking are therefore critical levers for improving advocacy.

02

Invest in Disruption Management and Service Recovery

Customer support failures, cancellations, and refund problems are the three most powerful predictors of non-recommendation, each associated with gaps of 15–31 percentage points. Faster resolution processes, proactive communication during disruption, and consistent compensation handling are higher-return investments than improving already-adequate service dimensions.

03

Manage the Customer Journey as an Integrated System

Recommendation behaviour is outcome-driven. A single critical failure, such as a missed connection, unresolved complaint, or cancelled flight, can outweigh multiple positive service moments. Airlines should monitor the entire journey, from booking to post-flight support, and treat consistency and reliability as the foundation of customer loyalty strategy.

LIMITATIONS AND CAVEATS

These findings should be interpreted with the following considerations in mind:

- ▶ Self-selection bias: Passengers who have had strongly negative experiences are more likely to submit reviews, which may overstate the prevalence of service failures in the broader customer base.
- ▶ Temporal relevance: Passenger expectations shift over time and in response to external events. Models should be periodically recalibrated to remain accurate.
- ▶ Predictive vs diagnostic value: While structured ratings predict recommendation with very high accuracy, text analysis is more valuable for diagnosing the specific nature of service failures and guiding targeted improvements.
- ▶ Sentiment model limitations: Automated sentiment tools (including both VADER and RoBERTa) misclassify a meaningful proportion of ambiguous reviews, particularly where positive language describes a negative outcome, or where blame is attributed externally.